Page No.
Date : / /

Gradient Descent + Ridge Regression.

$$L = (XW - Y)^T (XW - Y) + \lambda W^T W$$

$$w_0 = w_0 - \eta \frac{\partial L}{\partial w_0}$$

$$w_{new} = w_{old} - \eta \boxed{\frac{\Delta L}{\Delta W}} \rightarrow \begin{bmatrix} \frac{\partial L}{\partial w_0} \\ \partial L / \partial w_1 \\ \vdots \\ \partial L / \partial w_n \end{bmatrix}$$

$$L = \frac{1}{2}(XW - Y)^T (XW - Y) + \frac{1}{2}\lambda . W^T W$$

multiply by $1/2$ for mathematical convenience.

$$\frac{\partial L}{\partial W} = \frac{1}{2}\left[2X^T X W - 2Y^T X \right] + \frac{1}{2} 2\lambda W.$$

$$= X^T X W - X^T Y + \lambda W$$

Initialize $W = [\overset{w_0}{0} \quad \overset{w_1}{1} \quad \overset{w_2}{1} \quad 1 \quad 1 \quad \cdots \quad \overset{w_D}{1}]$

$$W = W - \eta \frac{\partial L}{\partial W}$$

Page No.
Date: / /

reg = SGDRegressor(penalty = 'l2', max_iter = 500, eta0 = 0.1, learning_rate = 'constant', alpha = 0.001)

Question:

a) If $\lambda \to \infty$ [λ → higher value].

then $w = [w_0\ w_1\ w_2 \ldots w_n] \to 0$

Shrink in value of coefficients.

x_1	x_2	x_3
$w_1 = 1000$	10	1
10	1	0.5

$\lambda \to \infty$

Higher value of w, decreases faster than that of smaller value.

b) Effect in Bias-Variance trade off.

low alpha → over fitting

optimal alpha → better result.

higher alpha → underfitting.

Both variance & bias depend in λ.

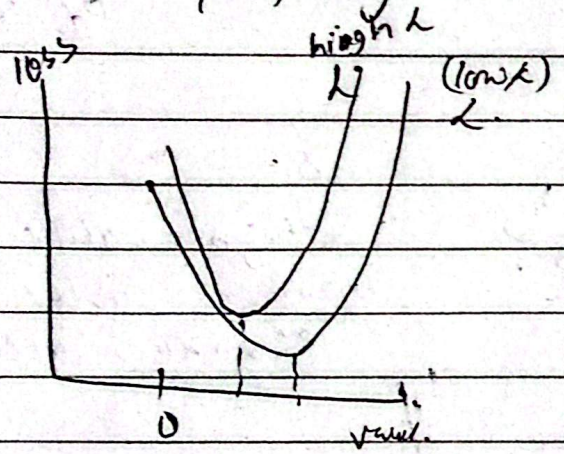
Page No.
Date : / /

c) Effect of λ in loss function.

$$L = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 + \lambda \cdot \|w\|^2$$

$\lambda \uparrow \rightarrow L \uparrow$

- We have to get better model by changing values of λ. While changing total loss increases, to deal with overfitting and make model works properly.

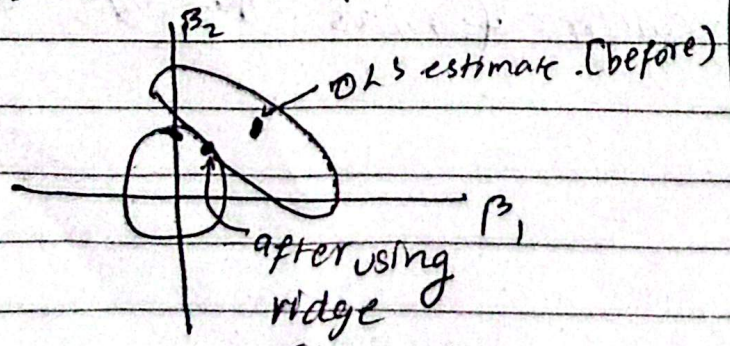


$\lambda \uparrow \rightarrow$ Loss curve shrinkage goes towards zero.

d) Why we called Ridge regression as Ridge regression?

$\rightarrow$ Loss = MSE + $\lambda \cdot \|w\|^2$

$$Loss = \sum_{i=1}^{n} (y_i - (\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2}))^2 + \lambda \cdot (\beta_1^2 + \beta_2^2)$$



minimum value will be in perimeter or periphery or ridge of circle.

So, ridge

{for now this is reason, later hard complex}